26900167 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECFS306 - MOBILE AND WIRELESS SECURITY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. State what eavesdropping is as a wireless attack. [K] / [CO1]
2. Identify some examples of Internet-enabled cell phones. [K] / [CO1]
3. Explain the purpose of information classification. [U] / [CO1]
4. Explain privacy in network security. [U] / [CO2]
5. Explain an Intrusion Detection System (IDS). [U] / [CO2]
6. Explain a security threat in wireless security analysis. [U] / [CO2]
7. Identify any two standards used in wireless security. [K] / [CO2]
8. Categorize IEEE 802.11 attacks based on the targeted layer. [An] / [CO3]
9. Subdivide the stages of WEP analysis. [An] / [CO3]
10. Distinguish between passive and active WEP attacks. [An] / [CO3]

11. Identify two factors that influence Bluetooth communication quality. [K] / [CO4]

12. Explain how device capability detection supports device-independent applications. [U] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain wardriving. [U] / [CO1]

14. Explain the concept of packet sniffing in wireless networks and discuss methods for protecting transmitted data. [U] / [CO2]

15. Explain the importance of maintaining and securely configuring the firmware of wireless hardware. [U] / [CO2]

16. Describe the functions of the monitoring, detection, and alerting components of an Intrusion Detection System. [U] / [CO2]

17. Explain why privacy protection is important in wireless communication. [U] / [CO2]

18. Explain how frame-control information is used to determine the type and purpose of a captured IEEE 802.11 frame. [U] / [CO3]

19. Examine the differences between IEEE 802.11 and 802.11b [An] / [CO3]

20. Examine the factors that influence the performance of satellite-based mobile communication systems. [An] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Describe WEP in detail. [U] / [CO1]

22. Discuss the various threats to wireless networks and explain suitable security mechanisms that can be used to protect against these threats. [U] / [CO2]

23. Examine how scalability can be achieved in pervasive web applications with a large number of connected devices. [An] / [CO4]

[10x2 = 20]